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Hericinoids A–C, cyathane diterpenoids from culture of mushroom *Hericium erinaceus*

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ABSTRACT

Three previously undescribed cyathane-type diterpenoids, hericinoids A-C (**1**–**3**) and three known analogues (**4**–**6**) have been isolated from fermentation broth of *Hericium erinaceus*. Their structures of hericinoids A-C were determined by in-depth analyses of NMR spectra and HRESIMS. The absolute configurations of compounds **1** and **2** were determined by the ROESY spectra along with the DP4+ calculations based on quantum chemistry. Compounds **1–5** were inactive on the enhancement of NGF-induced neutrite outgrowth in PC-12 cells. Instead, compounds **2**, **4**, and **5** showed potent cytotoxicity against HL-60 cell lines with the IC_{50} values of 18.3, 8.9, and 0.5 μM, respectively. Compounds **4** and **5** showed moderate cytotoxicities against MCF-7 cell lines with the IC_{50} values ranging from 13.4 to 15.8 μM.

1. Introduction

Hericium erinaceus (lion's mane mushroom), well-known as a tasty dish, has been attracting the attention of the world as there are various novel chemical constituents with versatile bioactivities (Friedman, 2015; Thongbai et al., 2015; Wu et al., 2015), among which isoindolinones (Chen et al., 2017; Wang et al., 2015a,b; Wang et al., 2016) and cyathane-type diterpenoids (Chen et al., 2016; Zhang et al., 2015) are conspicuous constituents. Up to date, numerous natural occurrences of cyathane-type diterpenoids have been discovered from the higher fungi (Chen and Liu, 2017; Enquist and Stoltz, 2009). One type of the cyathane-type compounds, generally consisting of annelated 5-6-7-membered ring attached with highly oxygenated pentose, mainly isolated from *Hericium* sp., *Cyathus* sp., and *Laxitextum* sp., has also gained considerable interest of the public for it possesses various pharmacological effects including pronounced neuroprotective activity, antibacterial and antifungal activity, and antitumor activities (Adeloju et al., 1996; Anke et al., 1977; He et al., 2016; Hecht et al., 1978; Huang et al., 2015; Kawagishi et al., 2006, 1994; Kenmoku et al., 2000, 2002; Lee et al., 2000; Ma et al., 2008; Mudalungu et al., 2016; Saito et al., 1998; Wang et al., 2014).

Expecting to discover unprecedented chemical structures with bioactivities, ongoing research on the mysterious mushroom is ecstatic especially by fermentation on the mushroom, which will broaden the chemical space of bioactive components. According to our detailed chemical study on the fermentation broth of *H. erinaceus*, three previously undescribed cyathane-type diterpenoids, hericinoids A-C (**1**–**3**) and three known analogues (**4**–**6**) were obtained. In this study, the absolute configurations of compounds **1** and **2** were assigned by deciphering ROESY correlations and DP4+ calculations based on DFT-quantum chemistry. Compounds **1–5** were evaluated the enhancement of NGF-induced neutrite outgrowth in PC-12 cells, but they were inactive. Cytotoxicities against five tumor cell lines (HL-60, SMMC-7721, A-549, MCF-7, and SW-480) were also evaluated.

2. Results and discussion

The fungal strain *H. erinaceus* was fermented under shaking condition at 22 °C for 25 days to obtain culture broth (50 L). The EtOAc extract (25 g) of the culture broth was subjected to the fractionation utilizing C18 medium performance liquid chromatography (MPLC) with a MeOH-H_2O gradient elution. Subsequent purification of interesting spots examined by anisaldehyde-sulfuric acid with Sephadex LH-20 column chromatography, silica gel column chromatography and

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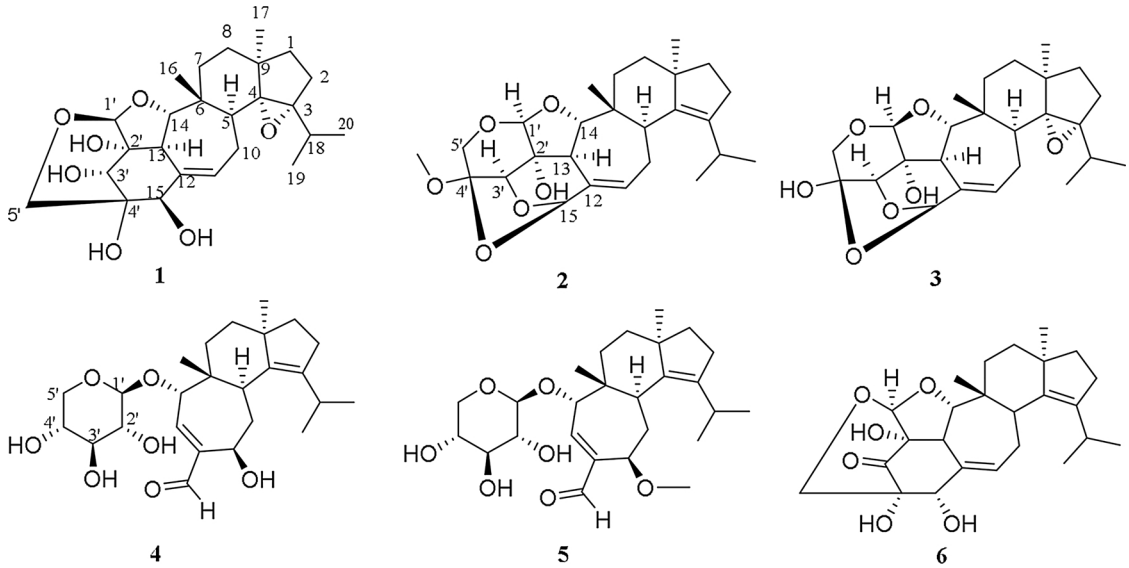

Fig. 1. Chemical structures of compounds **1–6** from fermentation broth of *Hericium erinaceus*.

eventually HPLC led to the isolation of compounds **1–6** (Fig. 1).

Compound **1** was obtained as white powder. Its molecular formula was determined to be $C_{25}H_{36}O_7$ with eight double-bond equivalents by sodium adduct (+)-HRESIMS ion peak at *m/z* 487.2092 $[M+Na]^+$ (calcd for 487.2093). The ^{13}C chemical shifts of compound **1** (Table 2) displayed 25 carbon atoms including four methyl, six methylene, eight methine and seven nonprotonated carbon signals. The olefinic carbon signals at δ_C 140.6 (s) and 127.6 (d) revealed one trisubstituted double bond in compound **1**. The ^1H NMR (Table 1) suggested two methyl doublets at δ_H 1.04 (d, J = 6.8 Hz) and 1.06 (d, J = 6.8 Hz) and two singlets at δ_H 1.03 and δ_H 1.14. The ^1H-^1H COSY correlations displayed the connectivity of five carbon-carbon bonds including H_2-1/H_2-2; H-5/H_2-10/H-11; H_2-7/H_2-8; H-13/H-14; H_3-19/H-18/H_3-20 (Fig. 2). The HMBC cross-peaks of H_2-1/C-3, H-5/C-4, C-6, H-11/C-13, H-15/C-11, C-12, H_3-16/C-5, C-6, C-7, C-14, H_3-17/C-1, C-4, C-7, C-8, C-9, H-18/C-4, H_3-19/C-3, C-18, C-20 suggested the existence of a fused 5/6/7

Table 1
^1H NMR Spectroscopic Data of Compounds **1–5.**

No.	1[a]	2[a]	3[b]	4[c]	5[d]
1	1.24 dd (13.0, 8.6)	1.65 ddd (14.7, 9.0, 5.7)	1.24 dd (12.3, 8.7)	1.64 ddd (14.1, 8.7, 5.7)	1.55 m
	1.17 ddd (13.0, 10.5, 8.5)	1.58 overlapped	1.15 ddd (12.3, 10.5, 8.5)	1.54 overlapped	1.52 m
2	1.85 ddd (14.1, 10.5, 8.6)	2.32 m	1.84 ddd (14.3, 10.5, 8.7)	2.33 overlapped	2.30 td (7.1, 1.9)
	1.77 dd (14.1, 8.5)		1.76 dd (14.3, 8.5)	2.29 overlapped	
5	2.57 d (11.5)	2.70 d (12.2)	2.57 d (12.4)	2.01 d (10.4)	3.40 dd (12.6, 1.9)
7	1.81 ddd (13.7, 13.7, 6.5)	1.70 td (13.4, 5.5)	1.70 ddd (14.5, 14.5, 4.3)	1.78 m	2.16 td (13.0, 4.7)
	1.46 ddd (13.7, 3.6, 3.6)	1.40 ddd (13.4, 4.4, 2.2)	1.42 ddd (14.5, 3.9, 3.9)	1.35 m	1.20 dt (13.0, 4.4)
8	1.69 ddd (13.7, 13.7, 3.6)	1.52 m	1.68 ddd (14.5, 14.5, 4.3)	1.54 overlapped	1.48 m
	1.39 ddd (13.7, 6.5, 3.6)		1.39 ddd (14.5, 4.3, 3.9)	1.43 m	1.44 m
10	2.35 brdd (18.0, 6.7)	2.62 ddd (17.5, 12.2, 2.4)	2.41 brdd (18.2, 6.9)	2.47 brdd (13.8, 9.0)	2.35 ddd (14.7, 5.6, 1.9)
	1.98 m	2.53 dd (17.5, 6.3)	2.04 ddd (18.2, 12.3, 4.3)	1.83 ddd (13.8, 10.4, 9.0)	2.05 overlapped
11	5.49 m	5.59 brdd (6.3, 2.4)	5.50 m	4.76 dd (9.0, 9.0)	4.47 brd (5.2)
13	3.16 d (10.8)	2.83 d (8.6)	2.82 d (8.8)	6.90 d (6.9)	7.08 d (7.5)
14	4.28 d (10.8)	4.13 d (8.6)	4.12 d (8.8)	4.51 d (6.9)	3.86 d (7.5)
15	4.73 brs	5.89 s	5.77 s	9.43 s	9.49 s
16	1.14 s	1.00 s	1.15 s	1.05 s	0.85 s
17	1.03 s	1.07 s	1.00 s	0.96 s	1.11 s
18	1.98 heptet (6.8)	2.89 heptet (6.8)	1.94 heptet (6.8)	2.90 heptet (6.8)	3.07 heptet (6.6)
19	1.04 d (6.8)	0.99 d (6.8)	1.06 d (6.8)	1.02 d (6.8)	0.99 d (6.6)
20	1.06 d (6.8)	1.01 d (6.8)	1.03 d (6.8)	1.01 d (6.8)	0.98 d (6.6)
1'	5.06 s	4.92 s	4.95 s	4.27 d (7.4)	4.72 d (2.4)
2'				3.24 dd (7.4, 7.8)	3.63 m
3'	3.76 t (1.7)	4.44 s	4.30 s	3.30 overlapped	3.59 m
4'				3.48 ddd (10.7, 8.9, 5.3)	3.56 m
5'	3.71 d (11.1)	3.85 d (12.8)	3.62 d (12.9)	3.81 dd (11.5, 5.3)	3.93 dd (12.5, 2.1)
	3.69 d (11.1)	3.60 d (12.8)	3.76 d (12.9)	3.15 dd (11.5, 10.7)	3.39 dd (12.5, 3.1)
11-OMe					3.24 s
4'-OMe		3.33 s			
2'-OH					4.60 d (5.4)
3'-OH					4.57 d (7.9)
4'-OH					4.19 d (11.0)

[a] Measured in methanol-d_4 at 500 MHz.
[b] Measured in methanol-d_4 at 800 MHz.
[c] Measured in methanol-d_4 at 600 MHz.
[d] Measured in acetone-d_6 at 500 MHz.

Table 2
^{13}C NMR Spectroscopic Data of Compounds **1–5**.

No.	**1**[a]	**2**[a]	**3**[b]	**4**[c]	**5**[d]
1	35.0	39.6	35.0	39.6	37.8
2	23.8	29.2	23.9	29.3	28.9
3	79.1	140.6	79.2	140.5	140.5
4	78.3	138.6	78.1	139.1	140.4
5	42.0	43.9	40.9	41.2	35.0
6	41.0	41.7	41.0	45.8	44.4
7	27.1	28.3	26.8	31.5	34.4
8	32.4	37.6	32.5	38.2	36.5
9	43.7	50.9	43.8	50.6	50.3
10	25.0	30.7	25.5	33.5	30.7
11	121.7	127.6	126.8	66.3	71.5
12	140.6	137.1	137.4	141.9	145.5
13	53.6	43.8	44.1	158.4	155.5
14	91.0	93.7	93.2	86.7	85.5
15	85.9	104.9	104.2	195.6	194.6
16	19.2	17.7	19.2	17.1	17.2
17	20.5	25.0	20.6	25.1	24.4
18	29.0	28.1	29.2	28.4	27.7
19	19.8	22.2	19.4	22.4	22.4
20	19.4	21.9	19.8	21.9	21.9
1'	108.5	106.2	106.5	108.0	105.3
2'	82.6	77.1	76.7	75.5	70.5
3'	71.0	84.8	86.3	78.2	70.4
4'	79.3	100.5	97.7	71.3	70.3
5'	64.3	63.8	64.5	67.2	62.0
11-OMe					57.2
4'-OMe		50.9			

[a] Measured in methanol-d_4 at 125 MHz.
[b] Measured in methanol-d_4 at 200 MHz.
[c] Measured in methanol-d_4 at 150 MHz.
[d] Measured in acetone-d_6 at 125 MHz.

tricarbocylic cyathane-type scaffold in compound **1**. The chemical shifts of C-11 (δ_H 5.49, δ_C 121.7) and C-12 (δ_C 140.6) were assigned to a double bond between C-11 and C-12, as supported by the ^{1}H-^{1}H COSY correlation of H-11/H-10 and the HMBC cross-peaks of H-15 with C-11, C-12 and C-13. Additionally, the proton of H-1′ at δ_H 5.06 (s) correlating to the carbon resonance at δ_C 108.5 revealed a ketal moiety that was formed by C-1 with C-14 and C-5′. The HMBC spectrum also denoted the correlations of H-15/C-3′, C-4′, H-1′/C-13, C-14, H-3′/C-2′, C-4′, C-5′, H-5′/C1′, C-4′, revealing the connection of C-1′/C-2′/C-3′/C4′/C-5′, which constructed the glycon as a highly oxygenated polycyclic pentose moiety, and also demonstrated the linkage of C-4′/C-15 and C-2′/C-13 with the help of HMBC correlations of H-14/C-2′, H-13/C-2′. Overall, the NMR data confirmed that compound **1** was resemble to CJ-15,544 (**6**) (Saito et al., 1998). The difference was that the carbonyl moiety (δ_C 207.6) in compound **6** was replaced by an oxygenated CH (C-3′, δ_C 71.0) in compound **1**. In addition, the double bond between C-3 and C-4 in **6** was replaced by an epoxy moiety (C-3, δ_C 79.1; C-4, δ_C 78.3) in **1**, as illustrated by the HMBC correlations of H_3-17/C-4 and H_3-18/C-3. As a result, the NMR data have determined the planar structure of compound **1**.

The configuration of this epoxy group was assigned on the following basis: (*i*) inspection of Drieding models strongly hinted that the *α*-oriented epoxy group in **1** gave less steric hinderance than that of *β*-oriented epoxy group, and oxidation of double bond was usually likely to occur from the less hindered side; (*ii*) the chemical shift of H_3-16 (δ_H 1.14) in **1** was considerably downfield relatively to the counterpart's methyl (δ_H 1.07) in **6**. These outcomes offered solid evidence of *α*-oriented epoxy group between C-3 and C-4 (Ayer et al., 1978). The ROESY spectrum recorded in acetone-d_6 showed the cross-peaks of H-13/H-5/H_3-17, OH-3′/H-13/H-15, indicating that H-5, H-13, H-15, H_3-17 and

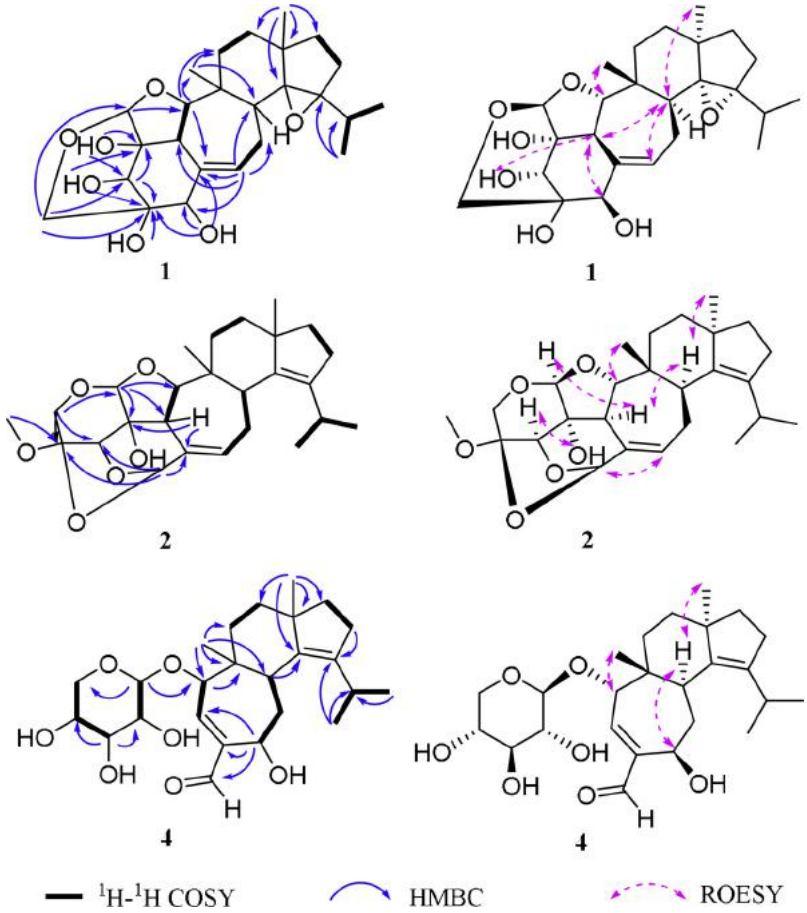

Fig. 2. Key COSY, HMBC and ROESY correlations of compounds **1**, **2**, and **4**.

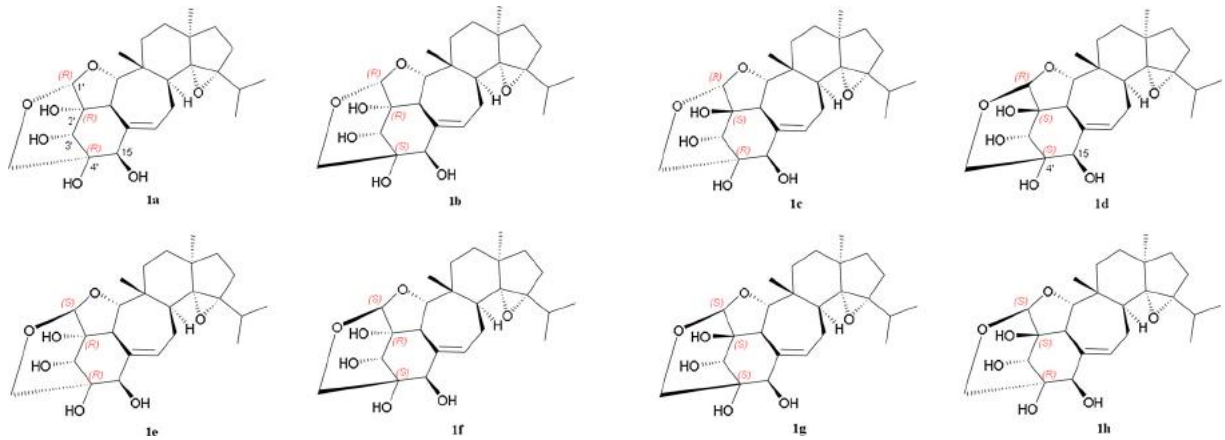

Fig. 3. 1a-1h: eight possible stereoisomers of **1**.

OH-3′ were *α*-oriented. Accordingly, the *β*-orientation of H-14 and H_3-16 was substantiated by the ROESY correlation of H-14/H_3-16. However, the stereochemistries at C-1′, C-2′ and C-4′ were still not assumed by the ROESY spectrum. The stereochemistry of **1** was finally determined by the consideration of the stability of **1** and its related ^{1}H-^{1}H correlations in the ROESY spectrum. Firstly, an exhaustive exploration of the stereochemistry at C-1′, C-2′ and C-4′ in **1** yielded eight possible candidate stereoisomers (Fig. 3, **1a–1h**). Preliminary conformational analyses denied **1a**, **1d**, **1e**, **1g** and **1h**, because **1a** defied the ROESY correlations of OH-3′/H-13 and H-15/H-13 in the ROESY spectrum of **1**, and the ball-stick models of **1d**, **1e**, **1g**, **1h** were unable to be established, indicating that they were not quite stable. The ROESY correlation of H-14/H-5′a, indicating the relatively close spatial distance between H-14 and H-5′a, defied **1b** and **1c**, because the spatial distances between H-14 and H-5′a in **1b** and **1c** were too far to correlate in the HMBC spectrum. Therefore, **1f** is the most likely candidate of compound **1**. Moreover, calculations of ^{1}H and ^{13}C NMR based on quantum chemistry is a way to recently pinpoint the stereochemistry of complex natural compounds (Lodewyk et al., 2012). The DP4 probability is an unbiased method designed to identify the correct structure among several possible candidates when only one set of experimental data is available (Grimblat and Sarotti, 2016; Grimblat et al., 2015; Smith and Goodman, 2010), as occurred in the case of compound **1**. The DP4+ probabilities of the possible stereochemistries for compound **1** were calculated (Table 3). The results also suggested that (3*R*, 4*R*, 5*S*, 6*R*, 9*S*, 13*R*, 14*S*, 15*R*, 1′*S*, 2′*R*, 3′*S*, 4′*S*)-**1f** was the most likely candidate, consistent with the outcome of the above stability analyses. Hence, the compound **1** was characterized and named as hericinoid A.

Compound **2** was also obtained as white powder. Its molecular formula $C_{26}H_{36}O_6$ with nine degrees of unsaturation was deduced by the (+)-HRESIMS ion peak at *m/z* 445.2577 $[M+H]^+$ (calcd for 445.2585). The ^{1}H and ^{13}C NMR data of compound **2** (Tables 1 and 2) showed 25 carbon atoms which were ascribable for five methyl, six methylene, eight methine (including two ketal groups of C-1′ (δ_C 106.2) and C-15 (δ_C 104.9)), and seven non-protonated carbon signals. The chemical shifts of C-11 at δ_C 127.6 accompanying with δ_H H-11 at 5.59, C-12 at δ_C 137.1, C-3 at δ_C 140.6, C-4 at δ_C 138.6 suggested the presence of two double bonds ($\Delta^{3,4}$ and Δ11,12). The ^{1}H-^{1}H COSY correlations of H_2-1/H_2-2; H-5/H_2-10/H-11; H_2-7/H_2-8; H-13/H-14; H_3-19/H-18/H_3-20, together with the HMBC correlations of H-5/C-4, C-6, C-11, H-11/C-13, C-15, H-13/C-12, H-14/C-12, H_3-16/C-5, C-6, C-7, C-14, H_3-17/C-1, C-4, C-8, C-9, H_3-19/C-3, C-18, C-20 suggested that compound **2** should also bear a tricarbocylic cyathane moiety. The HMBC correlations from δ_H 4.95 (H-1′) to δ_C 93.2 (C-14) and δ_C 64.5 (C-5′), as well as from δ_H 5.77 (H-15) to δ_C 86.3 (C-3′) and 97.7 (C-4′), suggested the existence of two ketal groups at C-1′ and C-15. Additionally, the HMBC correlations from H-1′ to C-13, C-14, C-2′, from H-3′ to C-2′, C-4′ and from H-5′ to C-3′, C-4′ manifested that the glycon was a highly oxygenated polycyclic pentose moiety. All information revealed that planar structure of compound **2** was closely similar to laxitextine B (Mudalungu et al., 2016) except for the presence of a methoxy group at C-4′ in **2** replacing the hydroxy in laxitextine B. In addition, C-3′ (δ_C 86.3, δ_H 4.30) in **2** was an oxygenated CH as elucidated by the HMBC analyses above. The HMBC correlations from the methoxyl group at δ_H 3.33 to C-4′ (δ_C 100.5) verified the presence of the methoxyl group at C-4′.

The stereochemistry of **2** was partially deduced by the ROESY experiment, as well as the comparison with that in laxitextine B (Mudalungu et al., 2016). Protons of H-5, H-13 and H-17 were assigned as the *α* orientation, while H-14 and H_3-16 were assigned as the *β* orientation according to the ROESY correlations. These data suggested that the aglycone part of **2** were intact cyathane diterpenoid. Moreover, the cross peaks of H-1′/H-5a/H-3′/OH-2′ in the ROESY spectrum in acetone-d_6 also suggested that H-1′, H-3′, OH-2′ were in the same side. The stereochemistries at C-1′, C-2′, C-4′ and C-15 were rationalized by calculations DP4+ probability. Firstly, an exhaustive exploration of the stereochemistry at C-1′, C-2′, C-3′, C-4′ and C-15 in **2** yielded eight possible candidate stereoisomers (Fig. 4, **2a–2h**). Preliminary conformational analyses got rid of five stereoisomers **2a**, **2c**, **2d**, **2f** and 2**h,** inasmuch as they were quite unstable. The remaining stereoisomers were calculated the DP4+ probability of **2b**, **2e** and **2 g**, respectively (Table 4). The results suggested that (5*R*,6*R*,9*R*,13*R*,14*S*,15*R*,1′*S*,2′*R*,3′*R*,4′*S*)-**2 g** was the most likely candidate. Therefore, the absolute configuration of **2** was determined and given the trivial name as hericinoid B.

Compound **3** was purified as white powder. Its molecular formula was determined to be $C_{25}H_{34}O_7$ with eight degrees of unsaturation by HRESIMS. The ^{1}H NMR and ^{13}C NMR data of compound **3** were nearly identical to those of compound **2**. The key difference was that an epoxy moiety between C-3 (δ_C 79.2) and C-4 (δ_C 78.1) in **3** replaced the double bond in **2**, as supported by the HMBC correlations of H_3-17/C-4 and H_3-18/C-3. In addition, C-4′ was substituted by a hydroxy group in

Table 3
DP4+ probability analysis of the stereoisomer **1b**, **1c**,**1e** and 1**f**.

probability	1b	1c	1e	1f
sDP4+ (H data)	0.00%	0.12%	0.06%	99.81%
sDP4+ (C data)	0.00%	0.01%	0.00%	99.99%
sDP4+ (all data)	0.00%	0.00%	0.00%	100.00%
uDP4+ (H data)	0.00%	0.13%	0.00%	99.86%
uDP4+ (C data)	1.30%	0.00%	98.36%	0.34%
uDP4+ (all data)	0.01%	0.00%	0.00%	99.99%
DP4+ (H data)	0.00%	0.00%	0.00%	100.00%
DP4+ (C data)	0.00%	0.00%	0.00%	100.00%
DP4+ (all data)	0.00%	0.00%	0.00%	100.00%

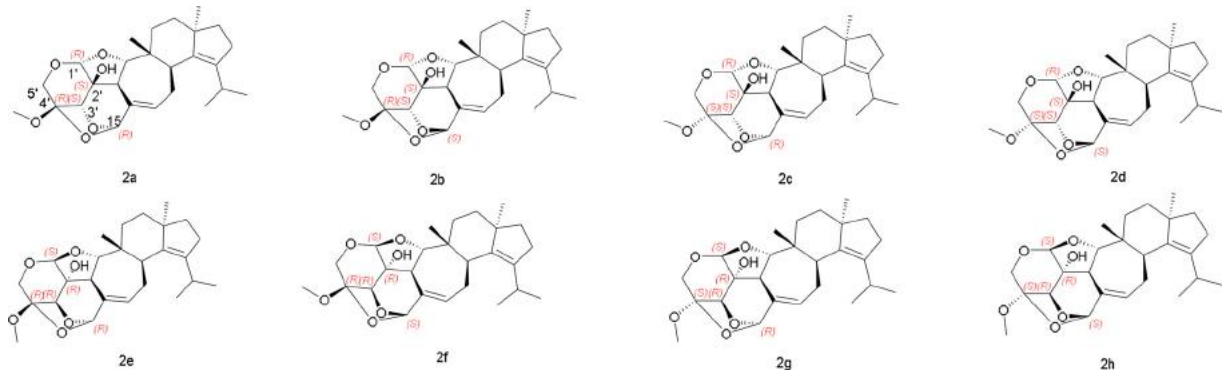

Fig. 4. 2a-2h: eight possible stereoisomers of **2**.

Table 4
DP4+ probability analysis of the stereoisomer **2b**, **2e** and **2g**.

probability	**2b**	**2e**	**2g**
sDP4+ (H data)	4.52%	0.00%	95.48%
sDP4+ (C data)	0.00%	0.00%	100.00%
sDP4+ (all data)	0.00%	0.00%	100.00%
uDP4+ (H data)	5.37%	0.00%	94.63%
uDP4+ (C data)	48.69%	46.18%	2.62%
uDP4+ (all data)	51.35%	0.00%	48.65%
DP4+ (H data)	0.27%	0.00%	99.73%
DP4+ (C data)	0.00%	0.00%	100.00%
DP4+ (all data)	0.00%	0.00%	100.00%

3 rather than the methoxy in **2**, as supported by the disappeared methoxy signals in the ^{1}H and ^{13}C NMR data. Detailed analysis of other NMR data suggested that the other parts of **3** were the same to those of **2**. The *β*-oriented epoxy group was deduced based on the Drieding models of **3** and chemical shifts of H_3-16 as exemplified by compound **1**. The relative configuration of other chiral centers were identified as the same to those in **2** by the ROESY data. Taken the co-isolation of **3** with **2** and biogenetic pathway into consideration, the absolute configuration of **3** was akin to that of **2**. The structure of **3** was thus assigned and named as hericinoid C.

Compounds **4** and **5** were isolated initially as two new natural products. However, during the process of this manuscript, the two compounds were just reported as erinacine Z1 and erinacine Z2, respectively (Rupcic et al., 2018). The spectroscopic data of compounds **4** and **5** are also included in the current report.

Compounds **1–5** were tested on their enhancement of NGF-induced neutrite outgrowth in PC-12 cells and also screened for cytotoxicity against five cancer cell lines HL-60 (human promyelocytic leukemia), SMMC-7721 (hepatic cancer), A-549 (lung cancer), MCF-7 (breast cancer), and SW-480 (colon cancer) using the MTS method, with cisplatin and paclitaxel as positive controls (Table 5). Compounds **1–5** showed no bioactivities on the enhancement of NGF-induced neutrite outgrowth in PC-12 cells (Figure S49-54). Intriguingly, compounds **2**, **4** and **5** showed potent cytotoxicity against HL-60 cell lines with the IC_{50} values of 18.3, 8.9, and 0.5 μM, respectively. Compounds **4** and **5** showed moderate cytotoxicity against MCF-7 cell lines with the IC_{50} values from 13.4 to 15.8 μM. The remaining compounds (**1** and **3**) were inactive cytotoxicity (IC_{50} > 40 μM) against five tumor cell lines and compounds (**2**, **4** and **5**) were inactive (IC_{50} > 40 μM) against the other three tumor cell lines.

3. Experimental

3.1. General experimental procedures

Optical rotations were recorded on a JASCO P-1020 digital polarimeter (Horiba, Kyoto, Japan). UV/Vis spectra were obtained using a Shimadzu UV2401PC spectrometer (Shimadzu, Kyoto, Japan). IR spectra were obtained using a Bruker Tensor 27 FT-IR spectrometer (Bruker Optics, Inc., Billerica, MA) with KBr pellets. 1D and 2D NMR spectra were measured on a Bruker Avance III 500 MHz, Bruker Avance III 600 MHz and Bruker Accend™ 800 MHz spectrometers (Bruker Biospin GmbH, Karlsruhe, Germany). HRESIMS were recorded on an Agilent 6200 Q-TOF MS system (Agilent Technologies, Santa Clara, CA, USA). Silica gel (200–300 mesh, Qingdao Haiyang Chemical Co., Ltd, P. R. China) and Sephadex LH-20 (Amersham Biosciences, Sweden) were used for CC. MPLC was performed on a Büchi Sepacore System equipped with pump manager C-615, pump modules C-605 and fraction collector C-660 (Büchi Labortechnik AG, Switzerland), and columns packed with Chromatorex C18 (40–75 mm, Fuji Silysia Chemical Ltd., Japan). Preparative HPLC was performed on an Agilent 1260 liquid chromatography system equipped with two types of Zorbax SB-C18 columns (9.4 mm × 150 mm and 21.2 mm × 150 mm, particle size 5 mm).

Table 5
Cytotoxicity IC_{50} values (*μ*M) of compounds **1–5** against Human Tumor cell lines.

compounds	Cell lines[a]	
	HL-60	MCF-7
1	> 40	> 40
2	18.3	> 40
3	> 40	> 40
4	8.9	13.4
5	0.5	15.8
cisplatin[b]	2.8	27.7
paclitaxel[b]	< 0.008	< 0.008

[a] Cell lines: HL-60, acute leukemia; MCF-7, breast cancer.
[b] Positive control.

3.2. Fungal material

The fungus *H. erinaceus* was collected from Shangri-La County, Yunnan province, China, in August 2007 and identified by Prof. Mu Zang at the Kunming Institute of Botany. The fungal strain used in this study was isolated from the fruiting bodies of *H. erinaceus*, which was also identified as *H. erinaceus* by partial sequence analysis of the internal transcribed spacers (ITS) sequence (Supplementary data). The BLAST query of this ITS sequence in NCBI (National Center of Biotechnology Information) database yielded *H. erinaceus* voucher ZJ2001HTJ02 (accession No. KU855351.1) as the closest match to the strain used in this study (99%).

3.3. Fermentation, extraction, and isolation

An *H. erinaceus* agar slant was transferred onto a PDA plate and incubated at 22 °C for 7 days. The agar plugs were cut into small pieces

(approximately 0.5 × 0.5 × 0.5 cm^3), then strain was inoculated into 170 Erlenmeyer flasks, respectively. Each 500 mL Erlenmeyer flask contained 300 mL of modified PDA medium (potato 200.0 g; glucose 20.0 g; KH_2PO_4 3.0 g; $MgSO_4$ 1.5 g and citric acid 0.1 g in 1 L of deionized water, pH was adjusted to 6.5 before autoclaving). Fermentation was carried out in two rotary shakers at 22 °C and 150 rpm for 24 days.

After the incubation, the whole culture was extracted with EtOAc. The crude extract (25 g) was fractionated by MPLC with a stepwise gradient MeOH/H_2O (v/v 20:80, 40:60, 60:40, 80:20, 100:0) to give five fractions (Fr.A–E). Fr.C (4.5 g) was subjected to MPLC with isocratic elution (MeOH/H_2O, 60:40) to obtain six subfractions (C1-C6) based on TLC analyses. Fraction C4 (1.5 g) was subjected to Sephadex LH-20 (MeOH) to get two fractions (Fr.C4a, Fr.C4b). Then Fr.C4b (986.5 mg) was purified over silica gel chromatography column, eluted with a stepwise gradient of petroleum ether/acetone (4:1-1:1) to get seven fractions (Fr.C4b.1-Fr.C4b.7). Compounds **1** (3.2 mg, t_R 5.0 min) and **2** (1.0 mg, t_R 9.7 min) was obtained from Fr.C4b.2 (302.4 mg) by prep-HPLC using a gradient elution (MeCN-H_2O 35%–45%, 7 mL min^{-1}, 20 min). Fr.C6 (0.9 g) was subjected to silica gel column with a gradient of petroleum ether/acetone (2.5:1–1.5:1) to get three subfractions (Fr.C6a, Fr.C6b, Fr.C6c). Compound **4** (1.8 mg, t_R 8.3 min) was isolated from Fr.C6c (105.2 mg) by HPLC, using a gradient elution (MeCN-H_2O 35%–45%, 7 mL min^{-1}, 20 min). Fr.D (4.3 g) was subjected to MPLC with isocratic elution (MeOH/H_2O, 80:20) to obtain two subfractions (D1 and D2) based on TLC analyses. D1 (2.5 g) was separated on silica gel chromatography column using the elution system of petroleum ether/acetone (3:1) to get four subfractions (D1a-D1d). Further purification of D1b (357.6 mg) was subjected to HPLC using a gradient elution (MeCN-H_2O 47%–62%, 7 mL min^{-1}, 30 min) to yield and compound **6** (13.5 mg, t_R 20.0 min). D2 was subjected to a silica gel column using a gradient of petroleum ether/acetone (4:1–1:1) to get six subfractions (D2a–D2f). Compound **3** (2.1 mg, t_R 11.8 min) was isolated from D2b (198.4 mg) by prep-HPLC using a gradient elution (MeCN-H_2O 55%–65%, 7 mL min^{-1}, 20 min). And compound **5** (1.9 mg, t_R 6.6 min) was isolated from D2d (165.6 mg) by prep-HPLC using a gradient elution (MeCN-H_2O 50%–60%, 7 mL min^{-1}, 20 min).

3.4. Spectroscopic data

3.4.1. Hericinoid A (1)

White powder; [α]20 D -42.0 (*c* 0.09, MeOH); UV (MeOH) λ_{max} (log ε) 204.5 (3.80) nm; IR (KBr) υ_{max} 3425, 2968, 2874, 1710, 1631, 1513, 1454, 1383, 1318, 1263, 1138, 1078, 1050, 1015, 890, 749, 595 cm^{-1}; ^1H and ^{13}C NMR data, see Tables 1, 2; HRESIMS: *m/z* 487.2092 [M + Na]$^+$ (calcd for $C_{25}H_{36}O_7Na$, 487.2093).

3.4.2. Hericinoid B (2)

White powder; [α]20 D -70.0 (*c* 0.05, MeOH); UV (MeOH) λ_{max} (log ε) 203.5 (3.89) nm; IR (KBr) υ_{max} 3438, 2934, 2865, 1613, 1457, 1381, 1339, 1187, 1091, 1069, 996, 909, 845, 796, 747 cm^{-1}; ^1H and ^{13}C NMR data, see Tables 1, 2; HRESIMS: *m/z* 445.2577 [M + H]$^+$ (calcd for $C_{26}H_{37}O_6$, 445.2585).

3.4.3. Hericinoid C (3)

White powder; [α]20 D -34.9 (*c* 0.04, MeOH); UV (MeOH) λ_{max} (log ε) 203.0 (3.33) nm; ^1H and ^{13}C NMR data, see Tables 1, 2; HRESIMS: *m/z* 485.1940 [M + K]$^+$ (calcd for $C_{25}H_{34}O_7K$, 485.1936).

3.4.4. Erinacine Z1 (4)

White powder; [α]20 D -24.7 (*c* 0.06, MeOH); UV (MeOH) λ_{max} (log ε) 202.6 (3.41) nm; IR (KBr) υ_{max} 3424, 2930, 2870, 1684, 1631, 1461, 1384, 1162, 1043 895, 605 cm^{-1}; ^1H and ^{13}C NMR data, see Tables 1, 2; HRESIMS: *m/z* 473.2500 [M + Na]$^+$ (calcd for $C_{25}H_{38}O_7Na$, 473.2510).

3.4.5. Erinacine Z2 (5)

White powder; [α]20 D -72.1 (*c* 0.26, MeOH); UV (MeOH) λ_{max} (log ε) 203.2 (3.99) nm; IR (KBr) υ_{max} 3427, 2959, 2932, 2871, 1692, 1630, 1457, 1384, 1088, 1066, 1026, 606 cm^{-1}; ^1H and ^{13}C NMR data, see Tables 1, 2; HRESIMS: *m/z* 487.2672 [M + Na]$^+$ (calcd for $C_{26}H_{40}O_7Na$, 487.2666).

3.5. Neurite outgrowth-promoting activity and cytotoxicity assay in vitro

The neurotrophic activities of the test compounds were examined according to an assay using PC12 cells as reported (Greene and Tischler, 1976). Briefly, PC12 cells were maintained in 1640 medium supplemented with 10% horse serum (HS), and 5% fetal bovine serum (FBS), and incubated at 5% CO_2 and 37°C. Test compounds were dissolved in DMSO. For the neurite outgrowth-promoting activity bioassay, PC12 cells were seeded at a density of $5*10^4$ cells/mL in 48-well plate coated with poly-L-lysine. After 24 h, the medium was changed to that containing 10 μM of each test compound plus 5 ng/mL NGF, or various concentrations of NGF (50 ng/ml for the positive control, 5 ng/ml for the negative control). The final concentration of DMSO was 0.05%, and the same concentration of DMSO was added into the negative control. After 72 h incubation, the neurite outgrowth was assessed under a phase-contrast microscope. Neurite processes with a length equal to or greater than the diameter of the neuron cell body were scored as neurite bearing cells. The ratio of the neurite-bearing cells to total cells (with at least 100 cells examined/view area; 5 viewing area/well) was determined and expressed as a percentage.

The human tumor cell lines HL-60, SMMC-7721, A-549, MCF-7, and SW-480 were used in the cytotoxic assay. These cell lines were obtained from ATCC (Manassas, VA, USA). Cells were cultured in RMPI-1640 or DMEM medium (Biological Industries, Kibbutz Beit-Haemek, Israel) supplemented with 10% fetal bovine serum (Biological Industries) at 37 °C in a humidified atmosphere with 5% CO_2. The cytotoxicity assay was evaluated by the 3-(4,5-dimethylthiazol-2-yl)-5-(3-carboxymethoxy-phenyl)-2-(4-sulfophenyl)-2*H*-tetrazolium, inner salt (MTS) (Promega, Madison, WI, USA) assay (Cory et al., 1991). Briefly, cells were seeded into each well of a 96-well cell culture plate. After 12 h of incubation at 37 °C, the test compound (40 μM) was added. After incubated for 48 h, cells were subjected to the MTS assay. Compounds with a growth inhibition rate of 50% were further evaluated at concentrations of 0.064, 0.32, 1.6, 8, and 40 μM in triplicate, with cisplatin and paclitaxel (Sigma, St. Louis, MO, USA) as positive controls. The IC_{50} value of each compound was calculated with Reed and Muench's method (Reed and Muench, 1938).

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.phytol.2018.07.006.

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